

impossible to adequately survey a field as broad as AI in a short space. And finally, surveys of subfields and individual research thrusts are readily available.²⁸ Rather than anything survey-like, we present a thumbnail sketch that orients around four points of contrast between data-driven AI and LEIAs. The contrasts involve meaning, reasoning, applications, and explainability.

Meaning in data-driven AI Data-driven AI operates over uninterpreted words; it does not pursue the computation of meaning as understood by linguists and philosophers. When the terms *meaning* and *semantics* are used to describe data-driven systems, they are redefined according to what the technologies can actually supply.

Although data-driven systems do not compute full semantic and pragmatic analyses, significant effort has gone into computing semantic and pragmatic features that can improve system performance, such as semantic roles, coreference relations, temporal relations, and discourse relations.²⁹ In addition, the statistical approach called *distributional semantics* operationalizes the intuitions that “a word is characterized by the company it keeps” (Firth, 1957) and “words that occur in similar contexts tend to have similar meanings” (Turney & Pantel, 2010).³⁰ Distributional models are good at computing similarities between words. For example, they can establish that *cat* and *dog* are more similar to each other than either of these is to *airplane*, since *cat* and *dog* frequently co-occur with many of the same words: *fur*, *run*, *owner*, *play*. Moreover, statistical techniques like Pointwise Mutual Information can be used to detect that some words are more indicative of a word’s meaning than others. For example, whereas *fur* is characteristic of dogs, very frequent words like *the* or *has*, which often appear in texts with the word *dog*, are not especially characteristic of dogs. Although distributional semantics has proven useful for applications like document retrieval, it is not a comprehensive approach to computing meaning since it only considers the co-occurrence of words. Among the things it does not consider are:

- the ordering of the words, which can have profound semantic implications: *X attacked Y* versus *Y attacked X*;
- their compositionality, which is the extent to which the meaning of a group of words can be predicted by the meanings of each of the component words; for example, in most contexts, *The old man kicked the bucket* does not mean that he struck a cylindrical open container with his foot;³¹ and

- any of the hidden sources of meaning in language, such as ellipsis and implicature.

Reasoning in data-driven AI Since data-driven AI systems do not compute meaning, they cannot reason in humanlike ways, even though some create the illusion of reasoning—as when playing chess or doing machine translation. But what defines reasoning for machines? Some developers believe that the definition lies in benchmarks.³² That is, if one establishes a criterion for reasoning, and if a system fulfills that criterion, then it has reasoned.

At the time of writing, the assessment of reasoning is being rigorously discussed with respect to large language models (LLMs), which can generate text and respond to certain kinds of queries in humanlike ways.³³ Responses to this new technology by professionals and lay people abound.³⁴ With respect to LEIAs, two points about LLMs are important:

1. **LEIA development is not superseded or invalidated by LLM technologies.** Mahowald et al. (2023) explain LLM achievements and deficiencies in terms of the juxtaposition between (a) formal linguistic competence, which is the ability to generate text that sounds like the given language, and (b) functional competence, which is the ability to understand and use language in the world. They say, “LLMs show impressive (although imperfect) performance on tasks requiring formal linguistic competence, but fail on many tests requiring functional competence” (p. 1). Bubeck et al. (2023, pp. 93–94) list deficiencies of the LLM called GPT4: the model does not know when it is guessing; it makes up facts; it uses a very limited context; it cannot learn or update to changes in the environment; it cannot be personalized to users; it appears to suffer from cognitive biases in reasoning; and its post hoc explanations can range from useful to wrong or unrelated. So, while LLMs are undoubtedly an exciting breakthrough that promises significant technological advances in a variety of practical applications, they do not offer theoretical advances in understanding human cognition or in creating computational models of reasoning based on understanding. Instead, they generate plausible texts by computing the most probable continuation of a text on the basis of an enormous store of text examples.
2. **LLMs can serve LEIAs in similar ways as other data-driven tools.** LLM technology can and should be used to support computational cognitive

modeling. The formal, surface-level linguistic competence of LLMs opens up possibilities for multi-engine approaches to certain tasks undertaken by LEIAs. So far, we have experimented with their use in selecting among candidate sentences in language generation (section 4.3) and in preparing learning materials for LEIAs (section 7.1.3). It is very early days in the availability of LLMs, so we have only just begun to flesh out their potential utility for LEIA operation.

As Yogi Berra opined, it is difficult to make predictions, especially about the future. At present, deep reasoned assessments of the impact of deep neural network-based LLM technology on science are clearly premature. We intend to return to this topic in our future work.

Let us return to the matter of comparing reasoning by LEIAs with reasoning by data-driven systems such as LLMs. Data-driven systems do not reason in humanlike ways, so when they are asked to explain, they have to concoct something post hoc, which might be useful, wrong, or irrelevant.

Application areas Most data-driven AI applications address a single capability and can be referred to as silo systems. Readers are well acquainted with the successes of silo AI, which offers systems for web search, route guidance, object recognition, playing chess, making product recommendations, machine translation, and so on.

But while silo AI has spectacularly fulfilled some desiderata, others remain outstanding. For example, silo systems tend to suffer from rigidity. If an AI system is trained to play a game on a grid with a particular layout, if any aspect of that layout is changed, then the system needs to be retrained from scratch—one cannot just tell the system, “Play just as before but use the additional rows as well.” Another concern involves ceilings of results. Leaderboards in system competitions regularly show the winner’s score in the 70th or 80th percentile, at which point developers abandon the task because the preferred methods offer no inroads for improvement. Yet another issue that has not made it onto the agenda of mainstream AI is addressing the challenges of complex application areas, such as clinical medicine viewed holistically. Finally, the lack of explanatory power of data-driven AI has been recognized as a serious limitation to its adoption by end users in critical domains, which is the topic to which we now turn.

Explanation in data-driven AI Explanation is a problem for data-driven AI systems because their methods of computation are opaque even to